

Design Project 1 Weekly Progress Report

Design Project Short Title: 4SIGHT: Goat Pregnancy Detector

Design Project Technical Title: Design and Development of an Automated Concoction Machine with Environment-Controlled Fermentation Chamber at Sactuario Nature Farms

Date of Title Defense: __May 16__

Proponents (Family Name, Given Name, Middle Initial / Student ID No.) :

1. Briones, Gerard Patrick M. / 2023-140241
2. Trillanes, Christian Dave, C. / 2023-140189
3. Rizal, Joaquin Angelo D. / 2023140171
4. Ornedo, Angel David S./ 2021-140063

Adviser/Client: Sir. Leonardo A. Samaniego, Jr.

Panelists:

1. Sergio R. Peruda, Jr. (Chair)
2. Einstein D. Yong (Lead Panelist)
3. Sir. Stanley Glenn E. Brucal (Member 1)
4. Luigi Carlo M. De Jesus (Member 2)

SPECIFIC OBJECTIVES/ FUNCTION/ MEANS/ SCOPE AND DELIMITATION/ CONSTRAINTS	TYPE	WEEK NUMBER (inclusive dates)	NOTES
To implement an accurate classification model that monitors the behavioral patterns of goats.	SO-1		
To incorporate a precise detection system that identifies pregnancy indicators.	SO-2		
To integrate a responsive notification feature that informs farmers when a goat shows signs of pregnancy.	SO-3		
Capture Video	F-1		Done
Store Video Footage	F-2		Done
Extract Image	F-3		Done
Enhance Image	F-4		Done
Detect Goat	F-5		Done
Track Posture	F-6		

Classify Behavior	F-7		
Log Activity	F-8		Done
Send Notification	F-9		Done
Display Data	F-10		Done
CCTV Camera	M-1		Done
Computer	M-2		
Open CV	M-3		Done
CLAHE	M-4		Done
YOLO26	M-5		Done
ByteTrack	M-6		Done
MySQL	M-7		Done
Telegram Chat bot	M-8		Done
Custom Dashboard	M-9		Done
Include a video camera setup capable of continuously monitoring goats within a confined area, capturing behavioral activity data.	S-1		Done
Utilize a classification model to analyze goat behavior patterns captured through video surveillance to identify early signs of pregnancy.	S-2		
Provide automated notifications to farm staff when a goat is detected as possibly pregnant.	S-3		Done
Generate logs on captured behavioral indicators of pregnancy.	S-4		Done
Detection will be limited to native goats within the pen.	D-1		
The system will not factor in temperature when detecting goat pregnancy	D-2		Done
Notification latency is limited by the signal reception in the area.	D-3		Done
The system will limit its focus to identifying behavioral patterns of goats as follows: Walking Isolation Aggression Eating Inactivity	D-4		
The total expense of the system must not exceed PHP 15,000.	C-1		
This project must comply with the Philippine Animal Welfare Act of 1998 (RA 8485 as amended by RA 10631).	C-2		
This project will take place at Santuario Nature Farm (SNF).	C-3		
The project will specifically focus on native goats at SNF.	C-4		